Honors Geometry – 3.5 Practice **WORKED KEY**

Equations of Parallel and Perpendicular Lines • Section 3.5

Show your work. The work is the point, not the answer. Each part ends with a self-check, so use it before you move on.

Part A — Slope first.

Label (x_1, y_1) and (x_2, y_2), then subtract in the SAME order top and bottom. Ex: (1, −2), (5, 6) → m = (6 − (−2)) / (5 − 1) = 8/4 = 2

1. Find the slope of the line through (−3, 8) and (5, −4).

m = (____ − ____) / (____ − ____) = _____

→ **m = −3/2**

(−4 − 8) / (5 − (−3)) = −12/8

2. Find the slope of the line through (2, −7) and (2, 5).

m = (____ − ____) / (____ − ____) = _____

→ **undefined**

(5 − (−7)) / (2 − 2) = 12/0 → vertical

3. Find the slope of the line through (−6, 4) and (3, 4).

m = (____ − ____) / (____ − ____) = _____

→ **m = 0**

0/9 = 0 → horizontal

4. A line with slope −2/5 passes through (−1, 6) and (9, ____). Find the missing coordinate.

−2/5 = (____ − 6) / (9 − (−1))

→ **y = 2**

run 10 → rise −4 → 6 − 4 = 2

✓ **Check yourself:** Your four answers: one negative fraction, one undefined, one zero, one positive whole number. Two of anything? #2 and #3 are the usual suspects.

Part B — Parallel, perpendicular, coinciding, or neither?

Solve for y first. Same slope, different b → parallel. Same slope, same b → coinciding. Slopes multiply to −1 → perpendicular. Otherwise → neither. Vertical and horizontal lines are always perpendicular.

5. $y = \frac{3}{4}x - 2$ and $y = \frac{3}{4}x + 5$

m_1 = ____ m_2 = ____

→ **parallel**

both slopes ¾; b = −2 and 5

6. $y = -5x + 1$ and $x - 5y = 15$

rewrite: y = _____

→ **perpendicular**

$y = \frac{1}{5}x - 3$; $(-5)(\frac{1}{5}) = -1$

7. $2x + 3y = 12$ and $y = -\frac{2}{3}x + 4$

rewrite: y = _____

→ **coinciding**

$2x + 3y = 12 \rightarrow y = -\frac{2}{3}x + 4$: same slope, same b

8. $y = 2x + 7$ and $y = -2x + 7$

m_1 = ____ m_2 = ____

→ **neither**

slopes 2 and −2: product −4

9. $x = -4$ and $y = 9$

vertical or horizontal? _____ _____

→ **perpendicular**

vertical and horizontal

10. $6x - 2y = 8$ and $y = 3x + 1$

rewrite: y = _____

→ **parallel**

y = 3x − 4: same slope, different b

✓ **Check yourself:** Exactly two parallel, two perpendicular, one coinciding, one neither. If the tally is off, recheck the problem you did NOT rewrite into y = mx + b.

Part C — Write the equation.

Get the slope of the given line, adjust it (same for parallel, negative reciprocal for perpendicular), push it through the point, solve for y. Ex: through (3, 1) ∥ y = 2x − 9 → y − 1 = 2(x − 3) → y = 2x − 5

11. Through (4, −3), parallel to $y = \frac{1}{2}x + 6$.

→ **$y = \frac{1}{2}x - 5$ (S)**

$y + 3 = \frac{1}{2}(x - 4)$

12. Through (−6, 1), perpendicular to $y = 3x - 8$.

→ **$y = -\frac{1}{3}x - 1$ (L)**

m = −⅓: $y - 1 = -\frac{1}{3}(x + 6)$

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13. Through (2, 9), parallel to 4x + y = 5.

→ **y = −4x + 17 (O)**
4x + y = 5 → m = −4; y − 9 = −4(x − 2)

14. Through (−3, −2), perpendicular to 2x − 5y = 10.

→ **y = −5/2 x − 19/2 (P)**
m = 2/5 → m⊥ = −5/2; y + 2 = −5/2(x + 3)

15. Through (0, −7), parallel to y = −x + 4.

→ **y = −x − 7 (E)**
(0, −7) is the y-intercept, so b = −7

16. Through (5, 1), perpendicular to y = ⅘x − 3.

→ **y = −5/4 x + 29/4 (S)**
m⊥ = −5/4; y − 1 = −5/4(x − 5)

Answer Bank

y = −4x + 17 **(O)**	y = ½x + 5 **(T)**
y = ½x − 5 **(S)**	y = −x − 7 **(E)**
y = −3x − 1 **(A)**	y = −5/4 x + 29/4 **(S)**
y = −⅓x − 1 **(L)**	y = −4x + 9 **(R)**
y = −5/2 x − 19/2 **(P)**	

✓ **Check yourself:** Write the letter for #11 – 16 in order. They spell a word this section is all about. Three entries are decoys: land on one and that problem is wrong.

11	12	13	14	15	16
S	**L**	**O**	**P**	**E**	**S**

Part D — Push further.

Same three moves, hiding behind a sentence.

17. Error analysis. Jo says the line through (1, 4) perpendicular to y = 2x + 3 is y = 2x + 2. Name Jo's mistake in one sentence, then write the correct equation.

→ **y = −½x + 9/2**
Jo used the PARALLEL slope. Perpendicular needs −½: y − 4 = −½(x − 1)

18. Write the line through (−2, 7) that is parallel to the x-axis. Then write the line through (−2, 7) that is perpendicular to the x-axis.

→ **y = 7 and x = −2**
parallel to the x-axis = horizontal; perpendicular = vertical

19. Line a passes through (−1, 2) and (3, k). Line a is perpendicular to y = −2x + 1. Find k.

→ **k = 4**
m⊥ = ½: (k − 2)/4 = ½ → k − 2 = 2

20. Line m is y = ⅓x + 2. Line n is perpendicular to m and passes through (6, −1). Do m and n cross on the y-axis? Find where they cross and justify.

→ **No: they cross at (4.5, 3.5)**
n: y = −3x + 17; ⅓x + 2 = −3x + 17 → x = 4.5 (not 0)

✓ **Check yourself:** Add your k from #19 to the x-coordinate of the crossing point in #20. You should get 8.5. If not, redo #19 first. It's faster.